



Workshop 3 «Academic skills»

Design Science Research (DSR)

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RECAP - Overview of Scientific Research Methods



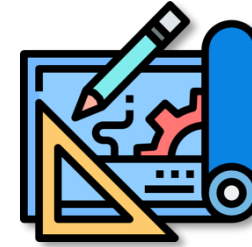
Qualitative Research

Focuses on understanding concepts, thoughts, or experiences and the context in which they occur.



Quantitative Research

Focuses on quantifying phenomena and relationships (that can be generalized to larger populations).



Design Science Research

Focuses on **creating** and evaluating artifacts (models, methods, systems) to solve identified problems.

Why Design Science Research?

In what situations is it used?

- ▶ When developing solutions to real-world problems that require innovation
- ▶ Ideal for fields like information systems, engineering, and technology, where practical applications are essential
- ▶ Useful when the goal is to create and evaluate new artifacts (models, methods, systems) to solve complex problems
- ▶ Employed in situations needing a rigorous approach to designing and testing new tools or frameworks

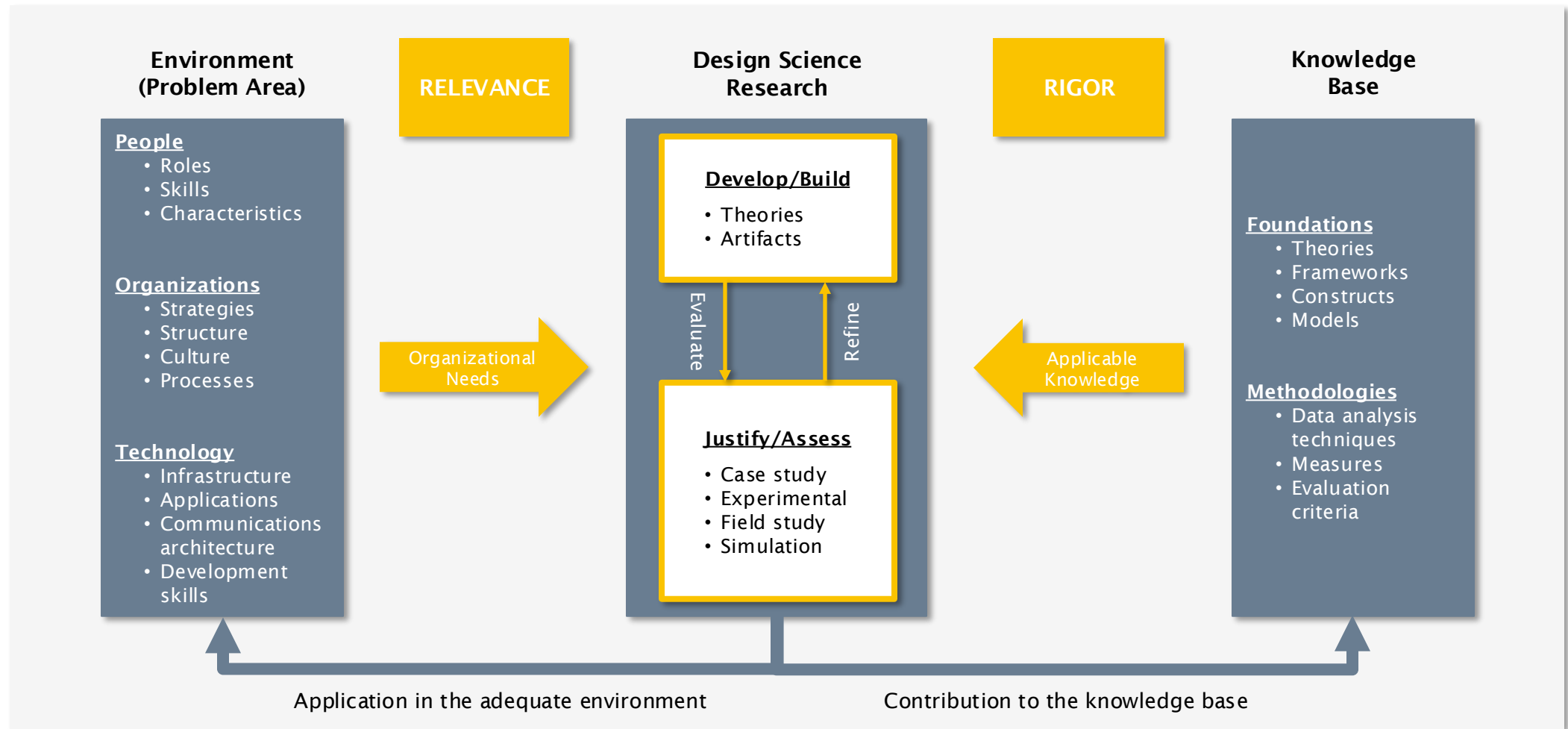
What is it used for?

- ▶ To **develop** new artifacts such as models, frameworks, processes, or systems that address specific issues
- ▶ To **improve** existing practices by applying a scientifically rigorous design process
- ▶ To **validate** solutions through iterative testing and evaluation, ensuring they meet intended goals
- ▶ To **bridge** the gap between theory and practice, linking academic research with practical, real-world solutions

What are its characteristics?

- ▶ **Iterative Process** - DSR involves cycles of design, development, and evaluation, refining the artifact with each iteration
- ▶ **Problem-Centered** - Focuses on understanding and solving real-world problems
- ▶ **Rigorous and Reproducible** - Combines creativity with scientific methods to ensure the solution's effectiveness and reliability
- ▶ **Evaluation-Focused** - Each artifact is tested to ensure it effectively addresses the problem
- ▶ **Knowledge Contribution** - Aims to produce both a practical artifact and theoretical insights that add to existing knowledge

The DSR Framework



Source: [Dresch, A., Lacerda, D.P., Antunes, J.A.V. \(2015\). Design Science Research](#)

DSR - Guidelines

Guideline	Description
GD 1: Design as an Artifact	Design-science research must produce a viable artifact in the form of a construct, a model, a method, or an instantiation.
GD 2: Problem Relevance	The objective of design-science research is to develop technology-based solutions to important and relevant business problems .
GD 3: Design Evaluation	The utility, quality, and efficacy of a design artifact must be rigorously demonstrated via well-executed evaluation methods
GD 4: Research Contributions	Effective design-science research must provide clear and verifiable contributions in the areas of the design artifact, design foundations, and/or design methodologies.
GD 5: Research Rigor	Design-science research relies upon the application of rigorous methods in both the construction and evaluation of the design artifact.
GD 6: Design as a Search Process	The search for an effective artifact requires utilizing available means to reach desired ends while satisfying laws in the problem environment.
GD 7: Communication of Research	Design-science research must be presented effectively both to technology-oriented as well as management-oriented audiences

Source: [Hevner et al., 2004](#)

DSR - Guidelines - Example

Example application of the guidelines for a new sales forecasting tool for restaurants based on supervised machine learning.

Guideline	Example
GD 1: Design as an Artifact	The result of the thesis will be an operational sales forecasting tool that can be used by Swiss restaurants to predict the expected sales in the upcoming week.
GD 2: Problem Relevance	Restaurants have to purchase foodstuffs and create work plans for employees at least one week in advance. The results of the sales forecasting tool will inform the planning to optimize purchasing and staffing decisions, ensuring sufficient availability while minimizing cost.
GD 3: Design Evaluation	The tool will be tested on historical sales data from multiple restaurants, comparing its predictions to actual outcomes to measure its accuracy and reliability.
GD 4: Research Contributions	The tool provides a novel application of supervised machine learning for sales forecasting in the restaurant industry. It will thus contribute to machine learning research and demonstrate the suitability of supervised machine learning for forecasting in the restaurant industry.
GD 5: Research Rigor	For tool development, I will analyze advanced supervised learning techniques, define restaurant requirements, and justify design decisions through a systematic suitability analysis. The tool's effectiveness will be tested in a real context using defined requirements and KPIs.
GD 6: Design as a Search Process	The development process involves exploring multiple clustering algorithms to identify the one that best meets the requirements for accuracy, scalability, and interpretability within the restaurant context.
GD 7: Communication of Research	The research findings will be presented in a report tailored for both technical audiences, detailing the algorithmic approach, and managerial audiences, emphasizing the tool's benefits for restaurant operations. Additionally, a management presentation is created.

DSR - Guidelines - Exercise (20 Minutes)

In this exercise, you'll work step-by-step through the DSR Guidelines to gain a foundational understanding of this research methodology. The goal of this exercise is to give you practical experience with the DSR methodology, providing you with a structured approach to conceptualizing and organizing research.

- ▶ **Choose a Topic:** Start by selecting a topic. If you already have an existing thesis topic, use that. If not, think of an area of interest you'd like to explore further.
- ▶ **Fill in the DSR Guidelines:** Using your topic, complete the DSR Guidelines. Take time to reflect on each guideline to ensure your responses align with the DSR methodology.
- ▶ **Review:** Once you've completed the table, review each section. Make sure that each part connects logically to the others, forming a coherent research structure.
- ▶ **Share it with a partner:** Share your guidelines with a partner, read it separately and give each other feedback. Guiding questions:
 - ▶ Did you understand everything?
 - ▶ Did the descriptions correspond to the guidelines?
 - ▶ Does everything logically build on each other?
 - ▶ Do you have advice for improvement?
- ▶ You can find the template for the DSR Guidelines table on Moodle under "Workshop 3 - DSR Guidelines Template.docx"
- ▶ Time: 15-20 min

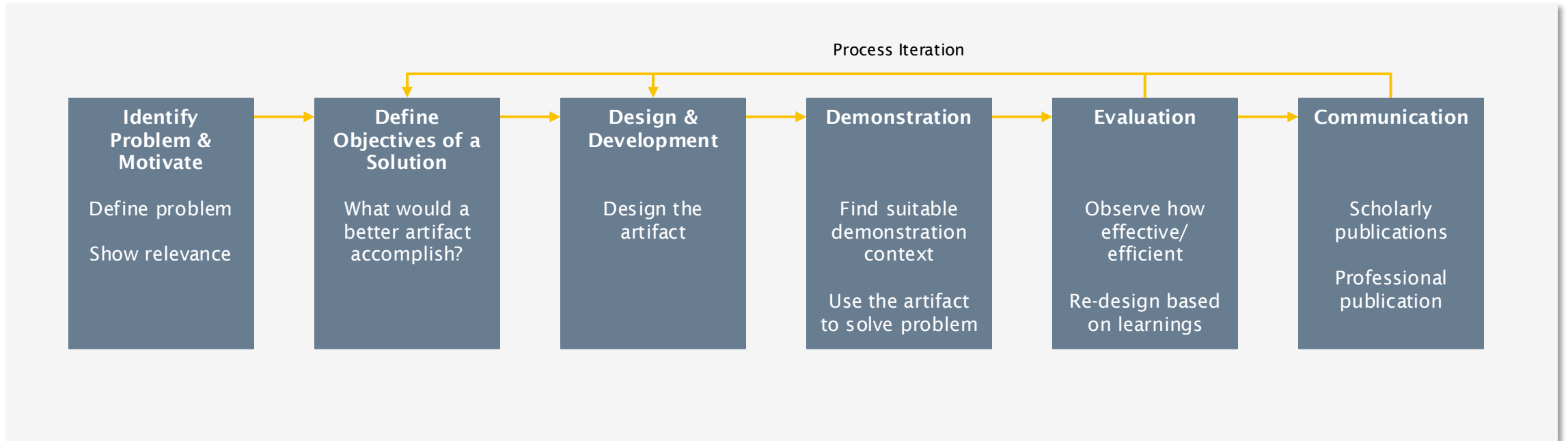
Workshop 3 – Design Science Research Guidelines

Topic:
Research Question:

Guideline	Description
GD 1: Design as an Artifact	
GD 2: Problem Relevance	
GD 3: Design Evaluation	
GD 4: Research Contributions	
GD 5: Research Rigor	
GD 6: Design as a Search Process	
GD 7: Communication of Research	

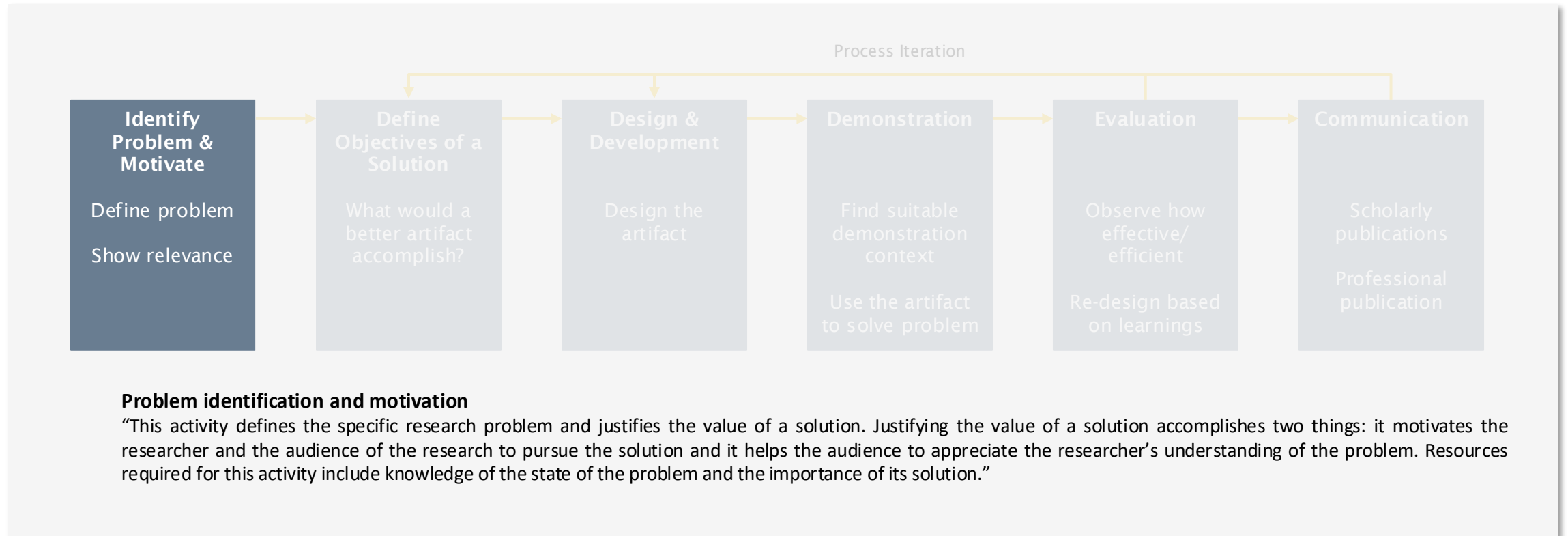
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DSR Methodology (DSRM) Process Model



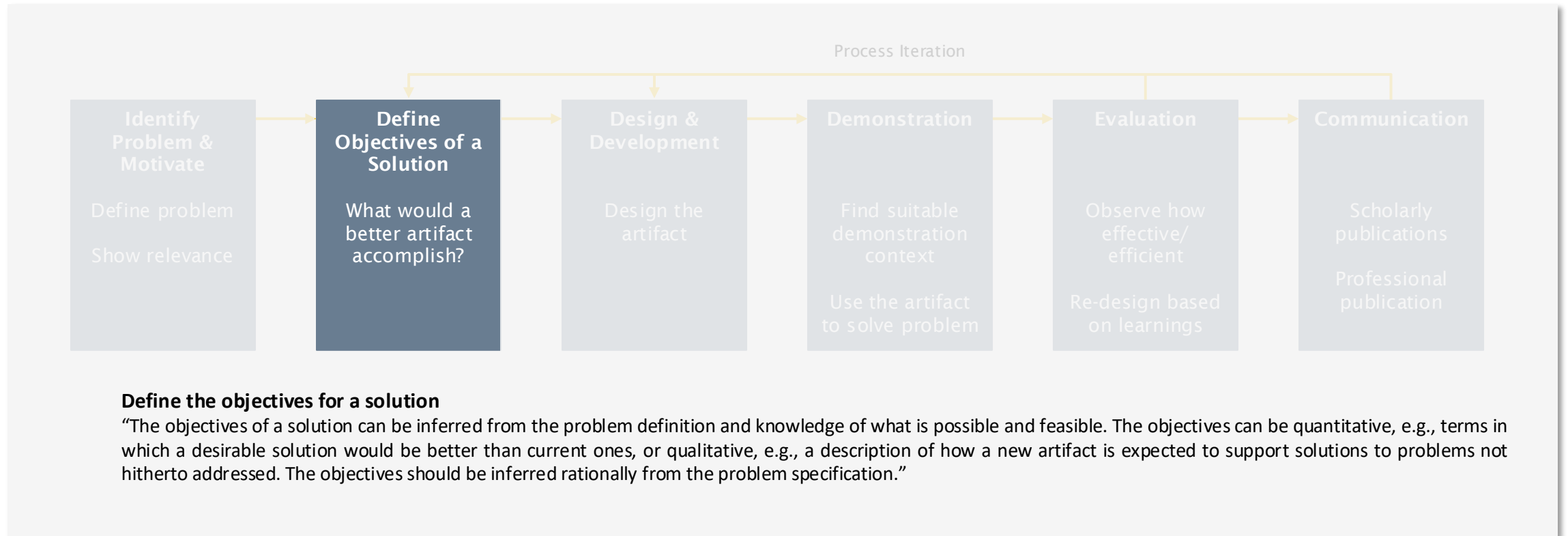
Source: [Vom Brocke, J., Hevner, A., & Maedche, A. \(2020\). Introduction to design science research. Design science research. Cases](#)

DSR Methodology (DSRM) Process Model



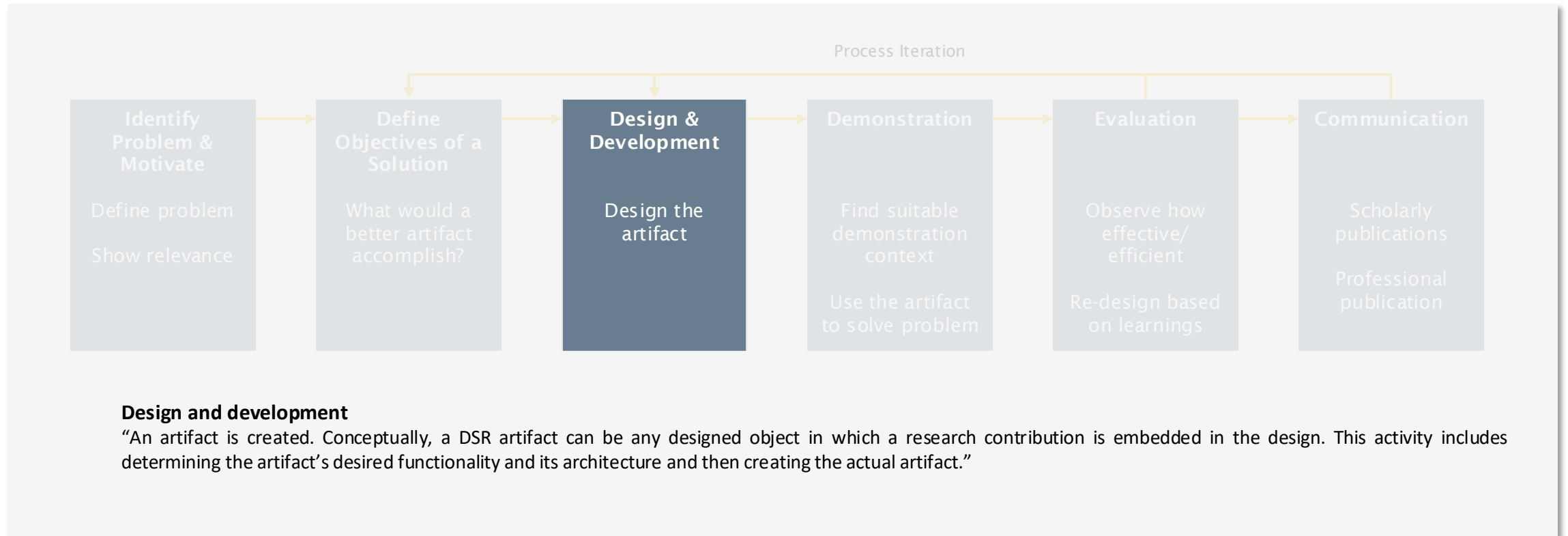
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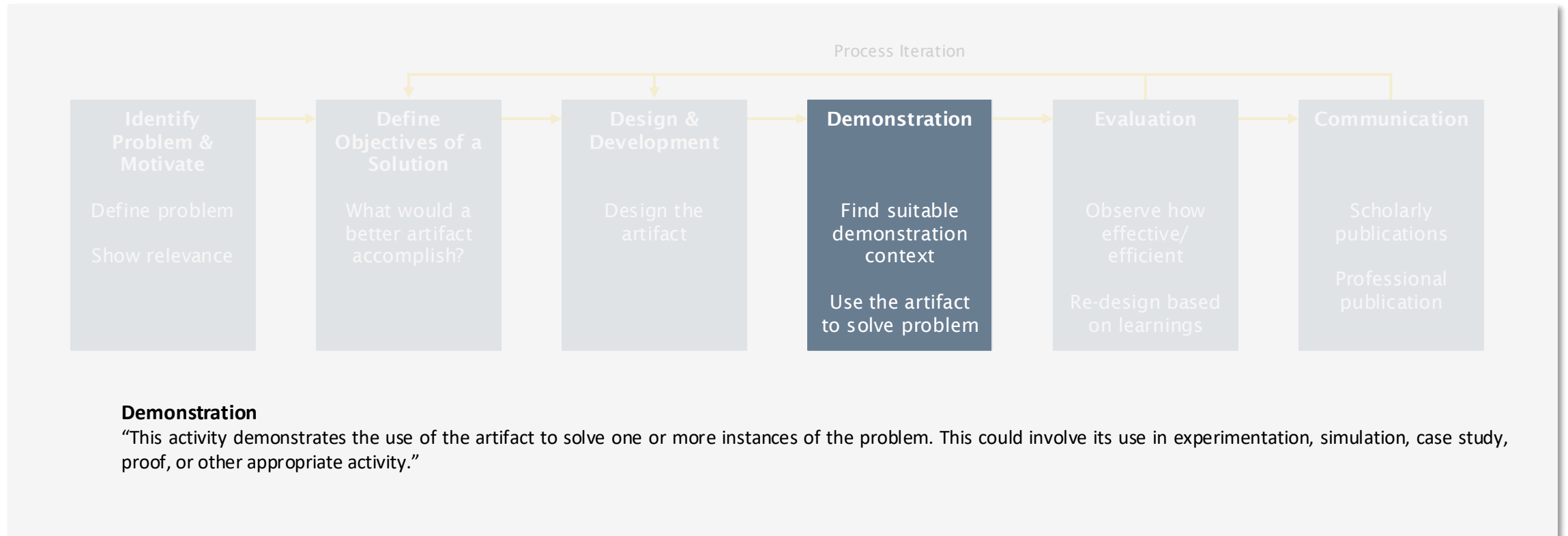
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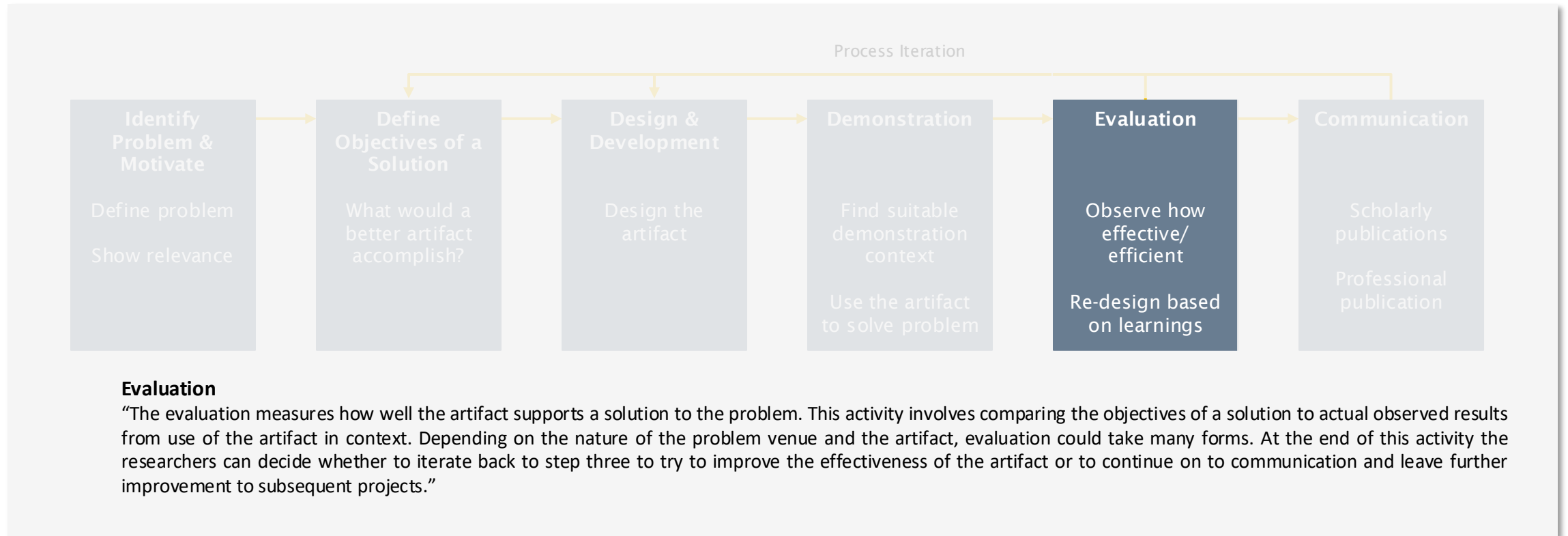
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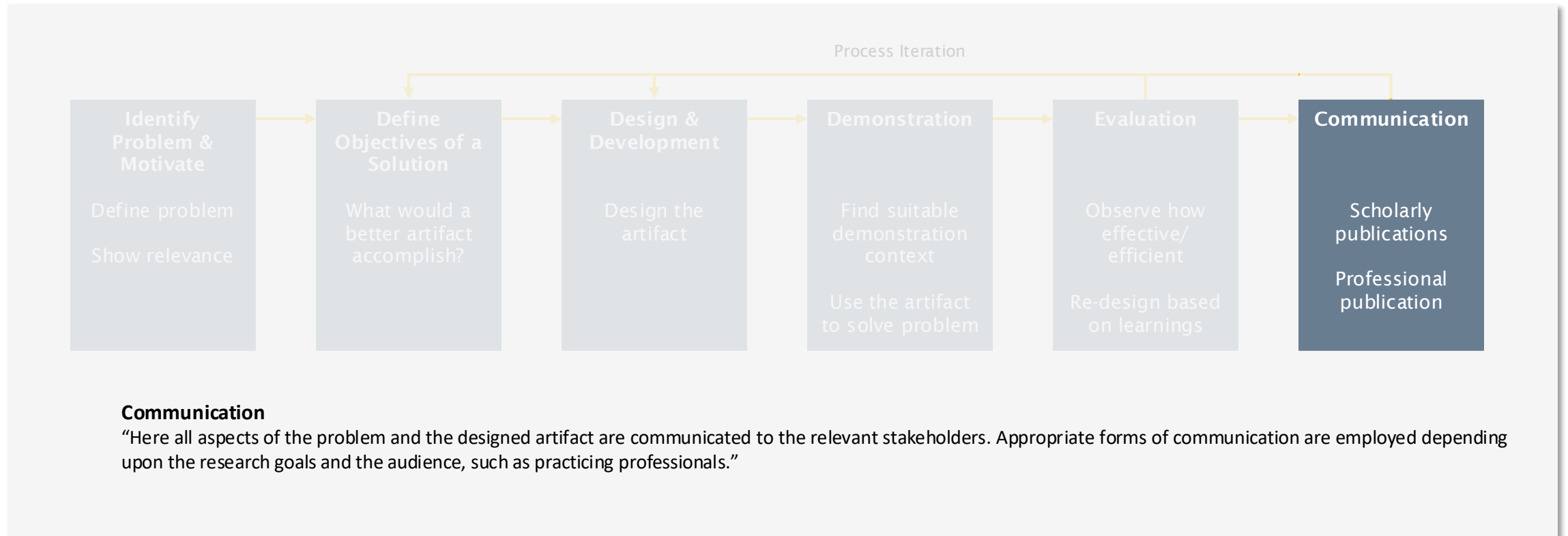
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DSR Thesis Structure and Content

n/a	Management summary
n/a	Table of contents
1	Introduction
2	Background and related work
3	Method
4	Requirements
5	Implementation
6	Evaluation
8	Discussion & Conclusion
n/a	Bibliography
n/a	Appendix

This is just an example structure, and you can adapt it to your thesis. It is recommended to align the structure with your supervisors.

DSR Thesis Structure and Content

Chapter content		Example (sales forecasting)
n/a	Management summary	▶ No example
n/a	Table of contents	
1	Introduction	
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n/a	Appendix	
▶ A good summary contains a brief description of the problem, the method, the results, conclusions as well as a brief outlook.		

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▶ The table of content shows the main chapters and sub-chapters. It is recommended to not show more than 2-3 hierarchies of chapters: ... 3 Method (hierarchy 1) 4 Results (hierarchy 1) 4.1 Sub-chapter (hierarchy 2) 4.1.1 Sub-chapter hierarchy (hierarchy 3) 4.1.1.1 Sub-chapter hierarchy (hierarchy 4) 5 Discussion (hierarchy 1) ... ▶ In general, it is not recommended to have more than 3-4 hierarchies in the text. Personally, I would not number a fourth hierarchy.		

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Chapter content

- ▶ A good introduction provides a clear context and formulation of the problem and explains why the problem is interesting.
- ▶ Usually, you start at a high abstraction level to embed the problem in a larger context and trigger the reader's interest. Then, you narrow it down to the particular problem you will be working on and show why it is relevant in the larger context.
- ▶ You can then clearly formulate the aim of the project and maybe even formulate research questions.
- ▶ In the introduction or method section, you should state what is in scope and what lies outside the scope of your work ("delimitation").

Example (sales forecasting)

- ▶ Purpose: Describe the need for sales forecasting in restaurants.
- ▶ Problem Statement: Highlight why forecasting sales is difficult (e.g., fluctuating demand, seasonality) and limitations of conventional forecasting methods.
- ▶ Proposed Solution: Introduce the idea of using supervised machine learning to overcome these challenges and why it is promising.
- ▶ Objective: State the main goal of the thesis (e.g., developing a tool that provides actionable insights for demand planning).
- ▶ Research Questions: Define specific questions the thesis aims to answer (e.g., "How accurately can supervised learning methods predict restaurant sales one week in advance?")
- ▶ Delimitation: Mention what is not in scope if required (e.g., other techniques might be promising, but I focus on supervised learning only)

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Chapter content

Context, existing research and research gap

- ▶ Describe in detail what is known about the problem context.
- ▶ Research and describe existing attempts (studies) that tried to address this or similar problems. What was their context, their approach and results?
- ▶ Describe potential gaps in the literature and why further research is needed (research gap). However, defining research gaps is optional in a bachelor thesis.

Techniques/methods/technologies

- ▶ Describe the key techniques/methods/technologies that are relevant for your project.

Example (sales forecasting)

2.1 Sales forecasting in restaurants

- ▶ Context: Provide a detailed explanation of forecasting challenges in the restaurant industry. If you have a concrete topic sponsor you can describe their specific context.
- ▶ Existing research: Discuss current forecasting approaches based on existing literature and their advantages and limitations.
- ▶ Research gap: Explain the need for a novel approach

2.2 Supervised machine learning for sales forecasting

- ▶ Supervised learning: Explain supervised learning, how it works and its potential benefits for sales forecasting.
- ▶ Techniques: Provide an overview of the key supervised learning techniques and their advantages/disadvantages.
- ▶ Approaches: Describe the key steps of a development process for supervised learning.

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Chapter content

Overall research approach

- ▶ Describe that you chose DSR and justify this decision.
- ▶ Describe your overall research approach, e.g., by illustrating the key steps in a figure and providing a short description of each step.
- ▶ You can use the DSR research methodology for that. But make sure to reflect on it and adapt it to your specific context if required.

Detailed research approach

- ▶ Describe each step in detail, the data and data sources used (e.g., interviews, databases) and the methods used (e.g. to analyze data, to choose suitable algorithms).

Example (sales forecasting)

2.1 Overall approach

- ▶ Describe that you apply DSR and explain why
- ▶ Summarize your overall approach (key steps), e.g. with a figure
- ▶ In the following, describe each step separately in more detail. Example:

2.2 Step 1: Requirement elicitation

- ▶ Describe how you identified the requirements (e.g. through interviews)

2.2 Step 2: Data collection

- ▶ Explain the process of gathering sales data from restaurants, including anonymized features like weather, holidays, and menu items.

2.3 Step 3: Algorithm Selection

- ▶ Describe how you compared and selected different algorithms (e.g. based on a value-benefit-analysis)

2.4 Step 4: Implementation

- ▶ Outline the steps to develop the algorithm and tool.

2.5 Step 5: Evaluation and refinement

- ▶ Detail how the tool will be tested (e.g., comparing its predictions with historical sales data) and the metrics used.

2.6 ...

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Chapter content

Requirements

- ▶ Provide an overview of the key requirements that you identified
- ▶ Describe each requirement
- ▶ Categorize and discuss the requirements according to suitable criteria, e.g.:
 - ▶ Functional vs. non-functional requirements
 - ▶ Must-have vs. optional requirements
 - ▶ System vs. user vs. business requirements

Example (sales forecasting)

2.1 Overview of requirements

- ▶ Provide an overview of the requirements, e.g. in a table.

2.2 Description of requirements

- ▶ Describe each requirement in detail and any important information about it (e.g. who postulated this requirement and why is it important).

2.3 Must-have vs. nice-to-have requirements

- ▶ Discuss which requirements are more important than others and why. Define which requirements have to be met for a successful outcome and which ones are rather nice-to-have.

2.4...

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Chapter content

Technical details

- ▶ Provide an overview of the architecture of your algorithm and tool.
- ▶ Describe the key components.
- ▶ Describe the key technologies used and explain why you chose them.

Visualizations

- ▶ Include screenshots and mockups of your tool if they are helpful.

Iterations

- ▶ Highlight key refinements made based on evaluations and feedback.

Example (sales forecasting)

2.1 Forecasting algorithm

- ▶ Describe which supervised learning algorithm you used and why you chose this one.
- ▶ If you did a systematic comparison of supervised learning algorithms, describe the results of this comparison (e.g. in a table).

2.2 Training and training data

- ▶ Describe the data used for training, how you trained the algorithm and stopping criteria.
- ▶ Describe whether you refined the data/algorithm and re-trained it based on intermediate results/evaluations/feedback (iterations).

2.3 Tool

- ▶ Describe how you developed a user-friendly tool for restaurant owners, describe its key functionality and optionally provide screenshots.
- ▶ Describe whether you improved the tool based on feedback (iterations).

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Chapter content

Evaluation

- ▶ Describe the results of your evaluation. The evaluation can be based on a systematic testing, interviews, simulations, comparison with other tools,....
- ▶ You can separately evaluate different components of the artifact (e.g. algorithm, tool).
- ▶ Also use the list of requirements defined earlier to check if they are met.
- ▶ Do not describe the evaluation process in detail, this should have been done in the method section.

Example (sales forecasting)

2.1 Forecasting accuracy

- ▶ Describe the results of the evaluation of the sales forecasting algorithm.
- ▶ Compare the evaluation results with those of conventional forecasting methods.

2.2 Tool usability

- ▶ Describe the results of the evaluation of the tool (e.g. usability based on user testing).

2.3 Requirements

- ▶ Describe which requirements are met and which ones not. Explain why.

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Chapter content

Results

- ▶ Here you describe your most important results, your practical and academic contribution

Discussion and limitations

- ▶ Here you can reflect on the validity of your results, e.g:
 - ▶ Is the available information sufficient for a conclusive answer to the questions? If not, why?
 - ▶ What are the unresolved challenges?
 - ▶ What other aspects that you have not considered still affect your question and how?
 - ▶ How do your results compare with the findings of other authors?

Future work

- ▶ Make suggestions for future work

Example (sales forecasting)

2.1 Overall results and answer to research questions

- ▶ Summarize the key insights derived from the evaluation of the algorithm and tool and explicitly answer the research question
- ▶ Highlight the practical contribution: Explain how the tool can improve restaurant operations (e.g., demand planning, inventory management).
- ▶ Highlight the academic contribution: Highlight contributions to academia (e.g. new insights on suitability of sales forecasting for restaurants)

2.2 Discussion and Limitations

- ▶ Compare your results to those of other studies or traditional forecasting methods.
- ▶ Describe why might your result not be completely valid? For example, there might be other algorithms or aspects that you could not test.

2.3 Future work

- ▶ Suggest areas for further research and improvements (e.g., incorporating external data like social media trends).

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▶ Include relevant auxiliary information in the appendix		
▶ If you have files, e.g. containing software code, please share them with your supervisors directly and upload them on Moodle		

DSR Thesis Structure – Exercise (20 Minutes)

Your Task

- ▶ Create a draft of your thesis structure:
 - ▶ Identify the main chapters, their subchapters, and provide 1-2 sentences describing the content of each. Specify roughly what you will write about (e.g. describe what concepts you will describe in the related work section)
 - ▶ Once you created your outline, read it again and make sure everything is aligned/logically builds on each other and relevant.
 - ▶ Share your structure with your peers and give each other feedback.
- ▶ Time: 25 minutes.
- ▶ You can find the template for the DSR Guidelines table on Moodle under “Workshop 3 – DSR Thesis Structure.docx”

Questions



Enjoy working on your Bachelor Thesis



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